A Null-Kinematic Reformulation of Spacetime, Gravity, and Quantum Mechanics

Toward a Unified Framework Based on Directional Indeterminacy

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Abstract

A reformulation of spacetime physics is proposed in which null propagation of physical influence at the speed of light constitutes the fundamental kinematic substrate, and spatial direction is not a primitive observable but an emergent, observation-defined quantity. In this framework, spacetime is treated as kinematically flat but dynamically flowing in its effective description and gravity arises from gradients and accelerations of this flow rather than intrinsic curvature. Special relativity, general relativity in the weak-field regime, and quantum mechanics emerge naturally from the same null-kinematic foundation. Quantum uncertainty is interpreted as a direct consequence of the fundamental unknowability of direction prior to observation. The framework reproduces known experimental results in the weak-field and nonrelativistic limits and suggests potential observational avenues distinguishing spacetime flow from curvature-based gravity. This reformulation provides a potential framework for progress toward a unified field theory. This work builds on established results in relativity and quantum mechanics while offering a reinterpretation of their physical content.

1 Introduction

Despite their empirical success, general relativity and quantum field theory remain conceptually incompatible. General relativity treats spacetime as a dynamical geometric object whose curvature encodes gravity[1], while quantum field theory presumes a fixed background and assigns primacy to quantum states and operators[2, 3]. This work adopts a different starting point: invariant null kinematic propagation at speed c, with spatial direction undefined prior to observation, is taken as the fundamental physical substrate. From this premise, spacetime, inertia, gravity, and quantum uncertainty emerge as secondary, relational phenomena. The principal physical significance of the null-kinematic reformulation lies not in modifying gravitational dynamics, but in enabling a natural and Lorentz-consistent coupling between spacetime structure and quantum phase evolution.

In conventional quantum field theory, physical descriptions are formulated within a chosen inertial reference frame, which is typically taken to be at rest without loss of generality. While this choice is mathematically valid, it implicitly privileges a frame in which the observer assigns zero spatial velocity to their local coordinates. The present reformulation challenges this interpretive convenience by treating invariant null motion as fundamental, such that no observer is ever truly at rest relative to massless degrees of freedom. This framework does not quantize spacetime, introduce additional particles, or alter the mathematical formalism of quantum mechanics or special relativity, but instead reinterprets their underlying kinematic primitives and physical content.

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This approach shares conceptual similarities with entropic gravity proposals [14], but focuses on kinematic directionality rather than thermodynamic entropy as the generative principle.

Mathematically, this formulation shares the structure of General Relativity expressed in Gullstrand–Painlevé coordinates [8, 9]; however, it reinterprets the associated spacetime flow [10] not as a mere coordinate artifact but as the underlying kinematic substrate. In this view, curvature emerges as a secondary description of flow gradients, while quantum mechanics arises from fundamental null-directional indeterminacy.

Throughout this work, the spacetime flow description is expected to be mathematically equivalent to the curvature formulation of General Relativity in all experimentally verified weak-field regimes. The distinction proposed here is ontological rather than algebraic: curvature is treated as an emergent geometric encoding of an underlying kinematic transport field. In standard regimes the flow and curvature descriptions are expected to be mathematically equivalent; any distinct predictions, if they exist, would arise only from additional structure beyond the mean flow (e.g., statistical degrees of freedom of the null ensemble), which is not fully developed here.

2 Logical Foundations

This work is motivated by the premise that any successful unification of gravity with quantum mechanics must rest on minimal and unavoidable physical principles, rather than additional structure. In this section, the foundational assumptions and logical constraints that motivate the spacetime flow formulation are presented.

2.1 Postulate 1: Universality of Relative Motion

Because massless particles admit no rest frame, no observer can be at rest with respect to them. Relative to massless propagation, every inertial observer is therefore characterized by motion at speed c, even though the direction of this motion is undefined prior to measurement.

2.2 Postulate 2: Direction Is Not a Primitive Observable

While the magnitude of relative velocity between an observer and a massless particle is fixed at c, the direction of this relative motion is undefined prior to measurement. This follows necessarily from: (i) Quantum uncertainty: Direction corresponds to momentum and unmeasured momentum is undefined; (ii) Isotropy of light propagation: Light propagates in all directions absent boundary conditions; and, (iii) Acceleration–direction equivalence: Because acceleration is frame-dependent, changes in motion cannot be uniquely decomposed into speed and direction in an observer-independent manner. In relativistic kinematics, and particularly in the limit of null propagation, acceleration alters the directional content of motion without changing its invariant magnitude. Direction therefore emerges as a relational property defined only through observation.

2.3 Postulate 3: Interpretations of the Equivalence Principle

The equivalence principle of general relativity admits two closely related but conceptually distinct interpretations: gravity may be described as intrinsic curvature of spacetime, or equivalently as accelerated spacetime motion producing non-inertial effects for embedded observers. This work adopts the latter interpretation and treats the equivalence principle literally, attributing gravitational phenomena to local acceleration of the spacetime flow rather than to curvature as a primitive.

In non-gravitational acceleration, such as an accelerating rocket, all external observers agree that apparent forces and observed light bending arise from the motion of the accelerating frame through otherwise flat spacetime, rather than from intrinsic curvature. By the equivalence principle, the same interpretive freedom applies to gravitational acceleration, motivating a description in which gravitational dynamics are attributed to accelerated spacetime flow toward mass concentrations. This description aligns with gravitational time dilation, redshift, light bending, and inertial frame dragging, without requiring intrinsic curvature.

2.4 Postulate 4: Null Motion Is Fundamental

Prior to directional resolution, the fundamental kinematic state is a null vector $\mathbf{k}$ constrained by magnitude:

$$|\mathbf{k}| = c. \tag{1}$$

The direction of propagation is represented by the unit vector $\hat{\mathbf{k}} \equiv \mathbf{k}/c$, distinct from the magnitude vector itself. Here c does not presuppose pre-existing spatial or temporal coordinates. It denotes an invariant propagation magnitude of the null substrate, from which the operational notions of spatial distance and temporal duration emerge only after directional resolution. Let $\mathbf{v}(\mathbf{x}, t)$ denote the emergent spacetime flow velocity field, and let $|\mathbf{v}| = \sqrt{\mathbf{v} \cdot \mathbf{v}}$ denote its Euclidean magnitude. Spacetime structure emerges from the statistical resolution of this null indeterminacy.

In this framework, every inertial observer is characterized by invariant motion at speed c prior to any decomposition into spatial or temporal components. Unlike the standard relativistic interpretation, this motion is not assumed to lie predominantly along a predefined time dimension. Instead, both spatial and temporal directions emerge only after directional resolution relative to observation. The null substrate introduced here is not a material medium with pressure, viscosity, or stress–energy of its own, but a kinematic structure describing the collective transport of unresolved null directions.

To be clear, directional indeterminacy is treated as ontic rather than epistemic. Ontic indeterminacy here refers to the absence of observer-independent directional structure, not to indeterminacy of physical existence. Prior to observation, motion is defined only by invariant null speed, while spatial direction has no observer-independent meaning. Direction emerges only through relational comparison between observer and system, not through revelation of a pre-existing value. This is analogous to relational descriptions in quantum mechanics, where momentum direction is undefined prior to measurement, but here it applies at the level of spacetime kinematics rather than wavefunction collapse.

Under these postulates, spacetime geometry is not fundamental but emergent. What is traditionally interpreted as curvature arises from spatial and temporal gradients in the spacetime velocity field. For example, as light passes near a massive object such as a star, it propagates locally in straight null trajectories relative to the surrounding spacetime flow[10]. The observed bending of light corresponds to the integrated path traced through a nonuniform, accelerating background. In this interpretation, what is commonly described as spacetime curvature is re-expressed as the cumulative effect of light propagating through a moving spacetime, without requiring curvature as a fundamental geometric primitive.

3 Null Directional States and Emergent Spacetime

This framework assumes that invariant null propagation at speed c constitutes the fundamental kinematic substrate of physical motion. Rather than assigning a primitive rest frame or intrinsic

spacetime geometry, physical states are described in terms of unresolved null-directional components whose resolution defines observed spacetime structure.

In one spatial dimension, null propagation admits two oppositely directed states. Denote these as right- and left-directed null components, each possessing invariant magnitude c. Prior to observation, these components are not individually resolved; only their collective structure is physically meaningful. Observable spatial motion arises from an imbalance in the directional weighting of these null states. Let w_+ and w_- denote the fractional contributions of the right- and left-directed null components, respectively. These weights satisfy the normalization condition

$$w_+ + w_- = 1 \tag{2}$$

reflecting conservation of the total null propagation capacity. The effective spacetime velocity state may then be expressed as a directional superposition:

$$\mathbf{v} = c(w_+\hat{\mathbf{n}}_+ + w_-\hat{\mathbf{n}}_-) \tag{3}$$

where $\hat{\mathbf{n}}_+$ and $\hat{\mathbf{n}}_-$ are unit vectors pointing in opposite spatial directions. Here, the quantity $\mathbf{v}$ represents the emergent observed velocity obtained by weighting the invariant null propagation $\mathbf{k}$ over unresolved directional components. In one spatial dimension, where $\hat{\mathbf{n}}_- = -\hat{\mathbf{n}}_+$, the observable scalar velocity reduces to

$$v = c(w_+ - w_-). \tag{4}$$

In the symmetric case $w_+ = w_-$, the observer assigns zero net spatial velocity, corresponding not to the absence of motion but to fully symmetric unresolved null propagation at speed c in opposing directions. As the directional imbalance increases, a greater fraction of the invariant null motion becomes resolved into a preferred spatial direction, producing a nonzero observed velocity. Thus, unlike conventional interpretations of special relativity, the state $v = 0$ does not represent spatial rest, but a maximally symmetric null-motion state.

The resolved spatial velocity v does not represent the magnitude of motion, which remains invariant at c, but rather the degree to which null motion has been directionally resolved relative to an observer's spatial coordinates. Even in the fully resolved limit, the invariant null propagation remains fixed at speed c. What changes is not the speed of propagation, but the degree to which this propagation is directionally resolved within spatial coordinates. The quantity v therefore represents an emergent spatial projection of null motion, not the physical transport speed of a particle or observer through space. Solving for the directional weights yields

$$w_\pm = \frac{1}{2}\left(1 \pm \frac{v}{c}\right). \tag{5}$$

The unresolved portion of null motion is quantified by a dimensionless indeterminacy parameter χ, defined as the geometric mean of the directional weights,

$$\chi = 2\sqrt{w_+ w_-} \tag{6}$$

which evaluates to

$$\chi = \sqrt{1 - v^2/c^2} \tag{7}$$

This quantity represents the fraction of null propagation that remains directionally unresolved. The Lorentz factor emerges naturally as the inverse of this unresolved fraction,

$$\gamma = \frac{1}{\chi} = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}. \tag{8}$$

In this interpretation, relativistic effects such as time dilation and length contraction arise not from intrinsic spacetime deformation, but from the redistribution of invariant null motion between resolved and unresolved directional components. As spatial velocity increases, the unresolved null capacity decreases, reducing the rate at which internal physical processes proceed relative to an external observer.

Physical states can then be represented as superpositions of null directional amplitudes over the unit sphere of propagation directions $\hat{\mathbf{k}}$:

$$\Psi = \int d\Omega_{\hat{k}} A(\hat{\mathbf{k}}) \, |\hat{\mathbf{k}}\rangle \, . \tag{9}$$

Here $d\Omega_{\hat{k}}$ denotes integration over the unit sphere of null propagation directions and $\hat{\mathbf{k}}$ denotes a unit direction on the null propagation sphere, not the magnitude-c null vector introduced earlier, with

$$\int |A(\hat{\mathbf{k}})|^2 \, d\Omega_{\hat{k}} = 1. \tag{10}$$

Spacetime emerges as a coarse-grained description of these null degrees of freedom. This distinction becomes essential in the quantum formulation developed later. Momentum and kinetic energy arise not from variation in propagation speed, but from the resolution of directional degrees of freedom within the invariant null substrate. The velocity field $\mathbf{v}$ should therefore be understood throughout this work as an emergent, observer-dependent quantity encoding directional bias, rather than a fundamental kinematic speed.

4 Spacetime Flow and Gravity

4.1 Spacetime Velocity Field

The spacetime flow velocity field $\mathbf{v}(\mathbf{x}, t)$ should be understood as the coarse-grained, macroscopic transport field arising from the underlying null-directional kinematics described in Section 3, where individual velocity states emerge from directional imbalance of invariant null motion. Hereafter, $\mathbf{v}(\mathbf{x}, t)$ refers exclusively to the emergent spacetime flow field, not the underlying null-speed constraint.

The gravitational acceleration field $\mathbf{g}$ is defined as the material acceleration of the spacetime flow:

$$\mathbf{g} = \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} \tag{11}$$

This field encodes the local transport of inertial structure relative to observers embedded within spacetime.

4.2 Effective Line Element of the Flow

The spacetime flow description admits an effective line element equivalent to General Relativity written in Painlevé–Gullstrand form. In terms of the flow velocity field $\mathbf{v}(\mathbf{x}, t)$, the metric can be written as

$$ds^2 = -c^2 dt^2 + (d\mathbf{x} - \mathbf{v}(\mathbf{x}, t)\, dt)^2 \, . \tag{12}$$

For spherically symmetric radial flow, this reduces to

$$ds^2 = -c^2 dt^2 + (dr - v(r)\, dt)^2 + r^2 d\Omega^2. \tag{13}$$

Mathematically, the effective metric possesses curvature whenever the flow has gradients; the claim of this work is ontological rather than mathematical—curvature is treated as a derived geometric encoding of a deeper kinematic transport field. This form makes explicit that null propagation occurs at speed c relative to the local spacetime flow.

4.3 Fundamental Field Equation

Mass acts as a source of spacetime acceleration:

$$\nabla \cdot \left(\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} \right) = -4\pi G\rho. \tag{14}$$

The divergence is selected because only the compressional component of spacetime flow contributes to volumetric convergence of inertial trajectories, directly reproducing Newtonian gravity. The sign convention is chosen such that the material acceleration of the spacetime flow is inward toward positive mass density. Since matter in free-fall is transported comoving with the local flow, this ensures that the observed gravitational acceleration $\mathbf{g}$ is attractive.

The present formulation is not asserted as a full replacement for Einstein's field equations, but rather reproduces Newtonian gravity in the weak-field limit. Recovery of post-Newtonian corrections is expected to arise from higher-order spacetime flow transport terms. Equation (14) is not presented as a complete covariant gravitational field equation, but as the minimal scalar sourcing relation required to recover the Newtonian limit within the flow interpretation. A fully tensorial, relativistically covariant generalization remains to be developed. Vorticity terms are not included at this level; rotational components are treated separately as higher-order or source-dependent corrections.

4.4 Newtonian Limit

For stationary, irrotational flow:

$$\Phi = -\frac{1}{2}|\mathbf{v}|^2 \implies \nabla^2\Phi = 4\pi G\rho. \tag{15}$$

In this formulation, the Newtonian gravitational potential corresponds to minus one-half the squared magnitude of the spacetime flow velocity, a relation formally analogous to Bernoulli's principle for a massless fluid. This ensures consistency with Poisson's equation and classical gravity.

4.5 Newtonian Limit as a Flow Toy Model

To demonstrate the calculational utility of the null-kinematic framework, consider a simple stationary, spherically symmetric flow corresponding to the Gullstrand–Painlevé form of the Schwarzschild solution. This choice of flow corresponds to a particular slicing of the Schwarzschild geometry and does not introduce new physics beyond the standard weak-field limit. In this regime, the spacetime flow is purely radial and stationary ($\partial\mathbf{v}/\partial t = 0$).

Following the Gullstrand–Painlevé structure, the radial flow velocity magnitude is defined as:

$$v(r) = -\sqrt{\frac{2GM}{r}} \tag{16}$$

where the negative sign denotes inward flow toward the mass center. For spherically symmetric radial flow, the scalar radial component is written as $v(r)$. In this framework, the observed gravitational acceleration a of a test particle at rest relative to the distant background is given by the material derivative of the flow field:

$$a = \frac{Dv}{Dt} = \frac{\partial v}{\partial t} + v\frac{\partial v}{\partial r}. \tag{17}$$

Substituting the stationary flow profile into the convective term:

$$a = \left(-\sqrt{\frac{2GM}{r}}\right)\frac{d}{dr}\left(-\sqrt{\frac{2GM}{r}}\right). \tag{18}$$

Evaluating the spatial gradient:

$$\frac{d}{dr}\left(-\sqrt{2GM}r^{-1/2}\right) = \frac{1}{2}\sqrt{2GM}r^{-3/2}. \tag{19}$$

Combining these terms yields:

$$a = -\sqrt{2GM}r^{-1/2}\left(\frac{1}{2}\sqrt{2GM}r^{-3/2}\right) = -\frac{GM}{r^2}. \tag{20}$$

This calculation demonstrates that the Newtonian inverse-square law emerges not as a static force or abstract geometric curvature, but as the material acceleration of a particle transported by gradients in the spacetime flow field. In this sense, Newtonian gravity admits a purely kinematic reinterpretation within the null-kinematic framework. This provides a transparent toy model showing how complex relativistic structures reduce to familiar classical results within the flow interpretation.

4.6 Motion of Light and Matter

Light propagation in this framework is described by the kinematic relation to spacetime as

$$\frac{d\mathbf{x}}{dt} = \mathbf{v} + c\,\hat{\mathbf{n}} \tag{21}$$

where light travels at invariant speed c relative to the local spacetime flow $\mathbf{v}(\mathbf{x}, t)$ [11], and $\hat{\mathbf{n}}$ denotes the locally resolved propagation direction. Apparent gravitational bending arises from spatial and temporal variations in the flow field rather than from intrinsic curvature. The observed trajectory of light reflects the superposition of invariant null propagation at speed c and transport by a locally moving spacetime background.

The motion of massive matter embedded in the spacetime flow is governed by the classical equation of motion

$$\frac{d^2\mathbf{x}}{dt^2} = \mathbf{g} \tag{22}$$

where the effective gravitational acceleration $\mathbf{g}$ arises from the local acceleration of the spacetime flow field. In this description, free-fall corresponds to motion comoving with spacetime, while non-inertial effects emerge from relative acceleration between matter and the flow. Light travels straight relative to spacetime; apparent curvature arises from spacetime flow gradients.

4.7 Gravitational Redshift as a Consistency Check

A necessary requirement of any relativistic gravitational framework is the recovery of gravitational redshift. Within the null-kinematic interpretation, this effect is not attributed to energy loss in a static potential or to intrinsic deformation of the time coordinate. Instead, gravitational redshift arises as a kinematic requirement for preserving invariant null propagation at speed c across regions of differing spacetime flow velocity. Because the null substrate itself is in motion, the frequency ν associated with a null-propagation process must adjust to remain consistent between regions characterized by different flow speeds $v(r)$. Consider a photon emitted at radius r_{em} and observed at radius r_{obs}. Here $v(r) = |\mathbf{v}(r)|$ denotes the magnitude of the radial spacetime flow speed. The observed frequency ratio is determined by the local flow states at emission and observation, yielding

$$\frac{\nu_{\text{obs}}}{\nu_{\text{em}}} = \frac{\sqrt{1 - \frac{v(r_{\text{em}})^2}{c^2}}}{\sqrt{1 - \frac{v(r_{\text{obs}})^2}{c^2}}}. \tag{23}$$

Using the stationary radial flow profile

$$|\mathbf{v}(r)| = \sqrt{\frac{2GM}{r}}. \tag{24}$$

Using the identification $\chi = \sqrt{1 - v^2/c^2}$ derived in Section 3, this expression becomes

$$\frac{\nu_{\text{obs}}}{\nu_{\text{em}}} = \frac{\sqrt{1 - \frac{2GM}{r_{\text{em}}c^2}}}{\sqrt{1 - \frac{2GM}{r_{\text{obs}}c^2}}}. \tag{25}$$

This reproduces the Schwarzschild gravitational redshift. Its emergence here demonstrates that relativistic frequency shifts are fully compatible with a kinematic spacetime-flow description and do not require intrinsic curvature of the temporal coordinate as a fundamental assumption.

4.8 Rotational Flow and the Lense–Thirring Analogue

The null-kinematic framework naturally generalizes to rotating systems by allowing the spacetime flow field $\mathbf{v}$ to possess vorticity [12]. For a central mass with angular momentum J, the flow acquires a transverse (azimuthal) component v_ϕ. In the weak-field limit, this component takes the approximate form:

$$v_\phi \approx \frac{2GJ}{c^2 r^2} \sin\theta. \tag{26}$$

The numerical prefactor here is given to leading order and may vary by factors of order unity depending on gauge choice and coordinate convention; the essential point is the emergence of a transverse spacetime flow proportional to angular momentum. Within this interpretation, the frame-dragging effects of general relativity arise from the physical vorticity of the spacetime flow itself. Rather than spacetime coordinates being "dragged" by a rotating metric, the null-propagation substrate undergoes rotational transport. Observable gravitomagnetic phenomena, such as gyroscope precession, are thus understood as kinematic consequences of spacetime vorticity. For stationary, axisymmetric solutions, the null-kinematic flow reproduces the Painlevé–Gullstrand form of the Kerr metric, with vorticity encoding frame dragging and angular momentum transport.

4.9 Equation of Motion and the Material Derivative

The dynamics of matter within the null-kinematic substrate are governed by the local acceleration of the spacetime flow. In this framework, the geodesic equation of general relativity is replaced by the material derivative of the flow field. The acceleration $\mathbf{a}$ of a test particle is given by

$$\mathbf{a} = \frac{D\mathbf{v}}{Dt} = \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v}. \tag{27}$$

This expression is identical to the gravitational acceleration field defined in Section 4.2, confirming that the motion of freely falling matter is governed by the same material acceleration that defines the spacetime flow dynamics. In this framework, the acceleration of the spacetime flow $\mathbf{a}$ and the observed gravitational acceleration $\mathbf{g}$ are kinematically equivalent for a particle in free-fall. Because the spacetime flow accelerates inward toward mass concentrations, freely falling matter is transported comoving with the local flow acceleration. Thus:

$$\mathbf{g} = \frac{D\mathbf{v}}{Dt} \tag{28}$$

where $\mathbf{g}$ represents the effective acceleration of a test particle as it follows the material acceleration of the null substrate. Equation (27) provides a unified description of inertial and gravitational motion. In a stationary radial flow, where $\partial \mathbf{v}/\partial t = 0$, the convective term $(\mathbf{v} \cdot \nabla)\mathbf{v}$ reproduces the Newtonian inverse-square law. In time-dependent or rotational configurations, the full material derivative captures the transport of a particle's state through a dynamically evolving spacetime flow.

Crucially, the flow field $\mathbf{v}$ does not represent a physical medium or preferred reference frame. It is a kinematic descriptor of null-propagation states. The material derivative thus serves as the fundamental bridge between spacetime flow dynamics and observed particle trajectories, distinguishing this framework from historical aether models. The flow field does not define a preferred inertial frame; all local observers resolve the same invariant null speed c, preserving local Lorentz invariance.

5 Quantum Mechanics from Null Directional Uncertainty

5.1 Uncertainty Relation

Directional indeterminacy has direct consequences for measurable conjugate quantities. In this framework, the invariant magnitude of null motion fixes the speed of propagation, while spatial direction remains unresolved prior to observation. As a result, increasing localization in position necessarily corresponds to a broader distribution over unresolved propagation direction. Because momentum is defined through directional resolution of null transport, this unavoidable tradeoff produces a lower bound on the simultaneous localization of position and momentum. The familiar Heisenberg relation

$$\Delta x \, \Delta p \gtrsim \hbar \tag{29}$$

can be interpreted as a kinematic constraint rather than an independent quantum postulate. Importantly, this same mechanism is consistent with special relativity: invariant null speed is preserved while directional information is redistributed, linking quantum uncertainty and relativistic invariance through a common underlying structure. In this sense, quantum uncertainty and relativistic kinematics are not separate principles but complementary manifestations of invariant null motion under incomplete directional resolution.

5.2 Schrödinger Equation

In this framework, mass is interpreted as corresponding to stored null anisotropy:

$$i\hbar\frac{\partial\Psi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\Psi. \tag{30}$$

Consequently, the Schrödinger equation can be reinterpreted as an effective description arising from unresolved null-directional motion within spacetime flow. Heuristically, isotropic coarse-graining of unresolved null directions leads to an effective diffusive operator, producing the Laplacian operator. The wavefunction Ψ does not represent a fundamental physical field, but a coarse-grained statistical encoding of direction-indeterminate null trajectories. Mass enters the equation as a measure of resistance to directional redistribution, governing the rate at which null motion spreads under coarse-graining. In simple terms, a persistent imbalance in null directional weights corresponds to energy that cannot propagate freely at speed c, giving rise to an effective rest mass that resists further directional redistribution. The familiar kinetic operator and time evolution emerge when invariant null speed is preserved while directional information is averaged over scales larger than those accessible to observation. In the appropriate nonrelativistic limit, this process is postulated to reproduce the standard Schrödinger equation, ensuring consistency with established quantum phenomena while providing a physical interpretation for quantum uncertainty, dispersion, and wave–particle duality. Quantum entanglement arises when multiple particles share a common unresolved null-directional structure established during interaction, prior to any observation that resolves directional information. Although such particles may separate in emergent spacetime coordinates, their joint quantum state remains non-factorizable until observation resolves directional information. Measurement of one particle updates the relational information available about the shared unresolved directional structure, producing correlated outcomes without introducing superluminal influence. This description is intended as an interpretive account consistent with standard quantum predictions, not as a modification of Bell-inequality–violating correlations. A fully relativistic treatment is expected to reproduce the Klein–Gordon and Dirac equations as higher-order descriptions of directional anisotropy dynamics; deriving this correspondence remains future work.

5.3 Gravitational Coupling

In this formulation, gravitational coupling enters quantum dynamics through spacetime flow rather than through an external potential or geometric curvature,

$$i\hbar\frac{\partial\Psi}{\partial t} = \frac{1}{2m}\left(-i\hbar\nabla - m\mathbf{v}\right)^2\Psi. \tag{31}$$

The modified kinetic operator, in which the momentum operator is shifted by the term $m\mathbf{v}$, analogous to the minimal coupling of a vector potential [13], reflects transport of the wavefunction relative to a locally moving spacetime background. This structure is mathematically equivalent to minimal coupling in conventional quantum mechanics, but the physical interpretation differs fundamentally: the velocity field $\mathbf{v}$ represents spacetime flow rather than a gauge potential. Mass appears as the coupling strength because it quantifies resistance to directional redistribution within null motion. In the absence of spacetime flow ($\mathbf{v} = 0$), this equation reduces to the standard free-particle Schrödinger equation:

$$i\hbar\frac{\partial\Psi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\Psi \tag{32}$$

ensuring consistency with established quantum mechanics. In the limit of weak and slowly varying flow, this equation reproduces standard gravitational phase shifts and free-fall behavior, while

providing a unified description of inertia, gravitation, and quantum transport as kinematic effects of spacetime motion.

Additionally, cosmic acceleration may arise from large-scale, time-dependent spacetime flow. In particular, the explicit time-derivative term in the spacetime acceleration equation, $-\partial \mathbf{v}/\partial t$, permits a small residual acceleration at cosmological scales if the background flow evolves slowly with time. In this interpretation, cosmic acceleration could be modeled as an emergent kinematic effect associated with slowly time-varying global transport, without introducing an explicit new component; a quantitative cosmological derivation is left to future work.

6 Discussion

The null-kinematic framework suggests that several structural features of modern physics may arise from kinematic constraints rather than independent postulates. In this picture, invariant null motion provides a conserved substrate, while observable spacetime structure emerges from the statistical resolution of direction. This shifts the role of geometry from fundamental to effective: curvature, inertial structure, and gravitational acceleration become large-scale descriptions of transport within an underlying flow.

6.1 The Emergence of 3+1 Spacetime Structure

Within the null-kinematic interpretation, the familiar 3+1 structure of spacetime is not a geometric background but a natural kinematic emergent. The invariant null substrate is fundamentally scalar, characterized intrinsically only by its propagation magnitude c. Spatial structure arises strictly through the process of *directional resolution*—the breaking of local symmetry in the substrate flow.

The three spatial dimensions appear to be the minimal isotropic vector space required to support continuous directional degrees of freedom and flux conservation. Time, conversely, is not a fourth geometric dimension but the scalar parameter quantifying the *unresolved* component of the null motion. The signature of the metric $(-,+,+,+)$ and the apparent asymmetry between space and time thus reflect their distinct kinematic roles: space as the domain of resolved flow, and time as the measure of residual propagation capacity.

6.2 Kinematic Origin of Quantum Uncertainty

This framework provides a kinematic origin for the Heisenberg uncertainty principle that is distinct from standard axiomatic quantum mechanics. Since the fundamental physical reality is null propagation with unresolved directionality, any attempt to localize a state in space requires a constraint on the directional weights $w_{\pm}$.

Because "position" is an emergent property of resolved flow and "momentum" is a measure of the flow's flux, simultaneous precision in both requires a simultaneous determination of the substrate's resolved and unresolved states—a kinematic impossibility. Consequently, quantum uncertainty is reframed not as an intrinsic fuzziness of nature, but as a statistical artifact of resolving a fundamental null motion into a fixed spatial basis. The Heisenberg limit ($\Delta x \Delta p \geq \hbar/2$) thus emerges as the lower bound of this directional resolution process.

6.3 Physical Interpretation of Mass and Energy

A primary insight of this reformulation is the physical origin of rest mass and the energy-mass equivalence. In Section 3, the state of spatial rest ($v = 0$) is shown to be a state of maximal

directional symmetry, where $w_+ = w_- = 1/2$. In this view, mass is not an intrinsic substance but represents invariant null motion that has been directionally balanced. The energy $E = mc^2$ thus corresponds to the total magnitude of the null substrate propagation required to maintain a spatially stationary state. The "conversion" of mass to energy is reinterpreted as the resolution of balanced null components into unidirectional propagation.

6.4 Singularity Avoidance and Information Conservation

The framework suggests a possible kinematic mechanism relevant to the black hole information paradox and the problem of infinite densities. As the spacetime flow velocity $\mathbf{v}$ approaches c near a mass concentration, the indeterminacy parameter $\chi = \sqrt{1 - v^2/c^2}$ approaches zero. Because χ represents the fraction of null propagation available for temporal evolution, the limit $v \to c$ defines a "kinematic saturation point."

At this boundary, the temporal degrees of freedom are fully resolved into spatial flow. This suggests that the interior of a black hole is not a region of infinite curvature, but a region of total directional resolution. Furthermore, because the null substrate satisfies the normalization condition $w_+ + w_- = 1$, the total information encoded in the null amplitudes $A(\hat{\mathbf{k}})$ is conserved. Within this interpretation, unitarity could be preserved because the substrate itself is treated as the fundamental conserved quantity, potentially preventing the "destruction" of information at a geometric singularity.

6.5 The Kinematic Arrow of Time

The "arrow of time" acquires a natural kinematic meaning. Time is reinterpreted as the scalar manifestation of the invariant null propagation speed c in the absence of total directional resolution. Because $|\mathbf{k}| = c$ is a fundamental constraint on the substrate (Postulate 4), physical processes must propagate. The irreversible nature of time follows from the fact that the null substrate cannot "un-propagate"; even at spatial rest, the substrate continues to move at speed c, producing the observed temporal flow.

6.6 Cosmological Implications: Primordial Saturation and Early SMBHs

The definition of the event horizon as a kinematic saturation boundary ($\chi \to 0$) offers a novel mechanism for black hole formation in the early universe. In the standard concordance cosmology, supermassive black holes (SMBHs) are presumed to grow from stellar-mass seeds via accretion; a process constrained by the Eddington limit. However, recent observations by the James Webb Space Telescope (JWST) reveal SMBHs at redshifts $z > 10$, with masses difficult to reconcile with standard accretion timescales.

In the null-kinematic framework, a horizon forms whenever the substrate flow velocity reaches saturation ($v = c$). During the radiation-dominated epoch, high-amplitude density fluctuations would induce rapid local accelerations in the substrate flow. Unlike stellar collapse, which requires hydrostatic equilibrium to be overcome, "direct kinematic saturation" can occur instantaneously in any region where the flow convergence satisfies $|\mathbf{v}| \to c$.

This raises the possibility that SMBHs could form directly from the primordial plasma via kinematic saturation, effectively "locking in" large horizons before the first stars ignited. These primordial saturation zones would act as gravitational seeds, accelerating structure formation and potentially explaining the presence of mature galaxies in the very early universe without violating the age constraints of the standard model.

6.7 Potential Extension to Gauge Interactions

The null-kinematic framework suggests a natural route toward incorporating gauge interactions without introducing additional fundamental fields. While gravity arises from gradients in the bulk transport velocity of the null substrate, gauge interactions may correspond to structured phase organization within the internal directional state space of unresolved null propagation.

In quantum mechanics, physical effects such as the Aharonov–Bohm phase shift demonstrate that the electromagnetic vector potential $\mathbf{A}$, rather than the magnetic field $\mathbf{B} = \nabla \times \mathbf{A}$, plays the fundamental role in quantum dynamics [13]. Within the present framework, this suggests that $\mathbf{A}$ may be interpreted as a kinematic phase-transport field defined over the same null-directional substrate that underlies spacetime flow. In this view, electric and magnetic fields arise as derived quantities describing gradients and vorticity of an underlying phase structure, analogous to how spacetime curvature emerges from gradients of the bulk flow velocity $\mathbf{v}$.

Charge, in this interpretation, would represent coupling to internal phase degrees of freedom of the null substrate, just as mass represents coupling to its bulk transport. This perspective naturally explains the gauge freedom of $\mathbf{A}$ as reflecting the unobservability of absolute phase within the unresolved directional ensemble. A full derivation of Maxwell's equations and non-Abelian generalizations from null-directional phase dynamics remains beyond the scope of the present work, but the structural parallels suggest a promising route toward unifying spacetime kinematics and gauge interactions within a single substrate-based framework.

Interpretive Note: It is important to emphasize that the present formulation does not deny the mathematical equivalence of curvature-based and flow-based descriptions within General Relativity in standard regimes. Rather, it proposes that the flow description may represent the more fundamental physical ontology, with curvature emerging as an effective geometric encoding of underlying kinematic transport. These interpretive features suggest a unifying substrate from which the mathematical structures of General Relativity and Quantum Mechanics may emerge as effective descriptions.

7 Observational Prospects and Possible Tests

The null-kinematic framework is constructed to reproduce the empirically verified predictions of General Relativity in all currently tested weak-field and stationary regimes. In these domains, the spacetime flow description is mathematically equivalent to curvature-based formulations (e.g., Painlevé–Gullstrand coordinates), differing primarily in physical interpretation rather than observable outcome.

Accordingly, the following discussion does not assume presently established deviations from classical GR. Instead, it identifies regimes in which the kinematic interpretation may provide either (i) alternative conceptual unification or (ii) possible future avenues for experimental discrimination if additional microphysical structure of the null substrate becomes dynamically relevant.

7.1 Quantum Transport in Nonuniform Spacetime Flow

In this formulation, gravitational coupling enters quantum dynamics through the transport operator

$$\hat{H} = \frac{1}{2m} \left(-i\hbar\nabla - m\mathbf{v}\right)^2 , \tag{33}$$

which is mathematically equivalent to minimal coupling in a metric theory expressed in flow form. In weak, slowly varying gravitational fields, this reproduces the standard gravitational phase shifts

predicted by conventional quantum mechanics in curved spacetime.

However, regimes in which the spatial extent of a quantum wavefunction becomes comparable to the characteristic gradient scale of the spacetime flow field $\mathbf{v}(x,t)$ may provide an interesting testing ground. In such cases, coherent quantum systems probe higher spatial derivatives of the background gravitational field. While these effects are expected to remain consistent with GR in the absence of additional substrate dynamics, large-baseline atom interferometers and macroscopic quantum superposition experiments offer increasing sensitivity to gravity-induced phase evolution in strongly inhomogeneous fields.

At present, the framework does not claim a confirmed deviation from metric gravity in these regimes; rather, it provides a unified transport interpretation of gravitational phase evolution that may prove useful if future precision measurements uncover anomalies beyond the standard post-Newtonian expansion.

7.2 Near-Horizon Kinematic Saturation

Within the null-kinematic interpretation, an event horizon corresponds to a saturation boundary where the spacetime flow velocity approaches the invariant null speed c. This reproduces the classical behavior of black hole horizons in General Relativity, including asymptotic freezing of outgoing null trajectories for distant observers.

The present formulation does not predict classical gravitational-wave echoes or horizon reflectivity in the absence of additional microphysical assumptions. Instead, the kinematic interpretation may offer conceptual clarity regarding trans-Planckian redshift, mode propagation near $v \to c$, and the interpretation of horizon thermodynamics. Any experimentally distinguishable deviation in this regime would require further development of the microscopic structure of the null substrate, which remains an open problem.

7.3 Cosmological Scaling Signatures

A more direct phenomenological consequence arises in cosmological settings. Because horizon formation corresponds to kinematic saturation, the framework naturally associates a characteristic mass scale with the cosmological expansion rate,

$$M_{\max}(z) \propto H(z)^{-1}. \tag{34}$$

If confirmed observationally, a redshift-dependent upper envelope for black hole masses that scales with the inverse Hubble rate would provide support for a saturation-based formation mechanism. The normalization factor in this scaling remains phenomenological at present, but the functional dependence constitutes the principal falsifiable cosmological prediction of the framework.

7.4 Interpretive vs. Dynamical Claims

It is important to distinguish interpretive reformulation from dynamical modification. In currently tested regimes, the null-kinematic theory is expected to remain observationally equivalent to General Relativity. Its primary contribution at this stage is conceptual unification: gravity as transport, quantum uncertainty as directional indeterminacy, and curvature as an emergent encoding of flow gradients.

Future work aimed at deriving the microphysical statistics of the null-directional substrate may determine whether additional scalar, vector, or higher-moment degrees of freedom become

experimentally accessible. Only such additional structure would lead to clear deviations from metric gravity.

Until such a derivation is completed, the null-kinematic framework should be understood as a reinterpretive unification with a small number of cosmologically testable scaling predictions, rather than as a presently established modification of General Relativity.

8 Consistency and Limits

The spacetime flow framework is constructed to remain consistent with all experimentally verified limits of relativistic and quantum physics. In the weak-field and stationary regimes, the theory reproduces the empirical predictions of general relativity, including gravitational time dilation, free-fall trajectories, and light deflection, through gradients and acceleration of the spacetime flow field. Local Lorentz invariance is preserved because all local observers resolve invariant null motion at speed c relative to the flow, ensuring agreement with special relativistic kinematics. Energy and momentum conservation are naturally associated with continuity of null transport and the absence of preferred frames. The framework is presently best understood as a reformulation that is empirically equivalent to GR in tested regimes; a primary open problem is identifying whether additional substrate degrees of freedom yield genuinely new observables. Instead, geometric effects emerge observationally from kinematic transport, allowing the theory to match established results. In regimes where General Relativity has been stringently tested, the spacetime flow description is expected to reduce exactly to an equivalent geometric formulation, differing only in interpretation rather than observable prediction. The present framework does not introduce new forces, additional propagating degrees of freedom, or deviations from classical GR predictions in tested regimes.

This work establishes the kinematic foundation for such a theory: while the spacetime flow field $\mathbf{v}$ corresponds to the first moment (dipole) of the null directional distribution, the second moment (quadrupole) of the same substrate naturally encodes transverse-traceless shear degrees of freedom, providing a kinematic origin for spin-2 gravitational waves. Whether this formulation reproduces all post-Newtonian and strong-field predictions of general relativity remains an open question requiring further development.

9 Conclusion

Null propagation and directional indeterminacy provide a unified foundation for spacetime, gravity, and quantum mechanics. Gravity emerges as spacetime flow; quantum uncertainty as null kinematics; special relativity as invariant null speed. This work is intended as a kinematic and interpretive foundation rather than a complete dynamical theory and is offered as a framework for further development rather than a finished unification.

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